Problem Set 2 Solutions

Turn in your solutions by Gradescope on the day they are due.

1. Using the definition of expectation, verify that for an n dimensional Gaussian random vector X with distribution $p(x) = \frac{1}{\sqrt{(2\pi)^n|\Sigma|}} e^{-1/2(x-\mu)^T\Sigma^{-1}(x-\mu)}$,

   (a) $E[X] = \mu$.
   Solution:
   Assume Σ to be positive definite.

$$\begin{aligned} E[X] &= \int xp(x)dx \\ &= \int x \frac{1}{\sqrt{(2\pi)^n|\Sigma|}} \exp\{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)\}dx \end{aligned}$$

   Using change of variable, $y = \Sigma^{-\frac{1}{2}}(x-\mu)$. So we have $dx = |\Sigma|^{\frac{1}{2}}dy$, $x = \Sigma^{\frac{1}{2}}y + \mu$.

$$\begin{aligned} E[X] &= \frac{1}{\sqrt{(2\pi)^n|\Sigma|}} \int (\Sigma^{\frac{1}{2}}y + \mu) \exp\{-\frac{1}{2}y^Ty\}|\Sigma|^{\frac{1}{2}}dy \\ &= \frac{1}{\sqrt{(2\pi)^n}} \int (\Sigma^{\frac{1}{2}}y) \exp\{-\frac{1}{2}y^Ty\}dy + \frac{1}{\sqrt{(2\pi)^n}} \int \mu \exp\{-\frac{1}{2}y^Ty\}dy \\ &= \frac{\Sigma^{\frac{1}{2}}}{\sqrt{(2\pi)^n}} \int y \exp\{-\frac{1}{2}y^Ty\}dy + \mu \int \frac{1}{\sqrt{(2\pi)^n}} \exp\{-\frac{1}{2}y^Ty\}dy \end{aligned}$$

   The first term is zero because the integrand is an odd function, the second term is μ because the integral equals to one (it is the integral of Gaussian PDF). Thereby, we have $E[X] = \mu$.

   (b) $E[(X-\mu)(X-\mu)^T] = \Sigma$.
   Solution:

$$\begin{aligned} E[(X-\mu)(X-\mu)^T] &= \int_x (x-\mu)(x-\mu)^T p(x)dx \\ &= \int_x xx^T p(x)dx - \mu\mu^T \end{aligned}$$

   Use the change of variables as in part (a):

$$\begin{aligned} \text{Cov(X)} &= \frac{1}{\sqrt{(2\pi)^n}} \int (\Sigma^{\frac{1}{2}}y + \mu)(\Sigma^{\frac{1}{2}}y + \mu)^T \exp(-\frac{1}{2}y^Ty)dy - \mu\mu^T \\ &= \frac{1}{\sqrt{(2\pi)^n}} \int \Sigma^{\frac{1}{2}}yy^T\Sigma^{\frac{1}{2}} \exp(\frac{1}{2}y^Ty)dy \\ &= \Sigma^{\frac{1}{2}} \frac{1}{\sqrt{(2\pi)^n}} \int yy^T \exp(-\frac{1}{2}y^Ty)dy\Sigma^{\frac{1}{2}} \end{aligned}$$

   Note that the integral in the middle is a matrix, off diagonal terms are zero because the integrand is an odd function. For the diagonal terms, take the first element as an example:

$$\begin{aligned} &\int y_1^2 \exp(-\frac{1}{2}y^Ty)dy \\ &= \int \cdots \int y_1^2 \exp(-\frac{1}{2}y^Ty)dy_1 \ldots dy_n \\ &= \int \exp(-\frac{1}{2}y_n^2) \cdots \int \exp(-\frac{1}{2}y_i^2) \cdots \int y_1^2 \exp(-\frac{1}{2}y_1^2)dy_1 \ldots dy_i \ldots dy_n \end{aligned}$$

Using integration by parts,

$$u = y_1, \qquad v = -\exp(-\frac{1}{2}y_1^2)$$
$$du = 1, \quad dv = y_1 \exp(-\frac{1}{2}y_1^2)$$
$$\int y_1^2 \exp(-\frac{1}{2}y_1^2)dy_1 = uv|_{-\infty}^{\infty} - \int v du = \sqrt{2\pi}$$

Thus we have $\int \exp(-\frac{1}{2}y_n^2) \cdots \int \exp(-\frac{1}{2}y_i^2) \cdots \int y_1^2 \exp(-\frac{1}{2}y_1^2)dy_1 \dots dy_i \dots dy_n = \sqrt{(2\pi)^n}$. Thereby $\text{Cov(X)} = \Sigma^{\frac{1}{2}}\Sigma^{\frac{1}{2}} = \Sigma$.

2. Show the following:

   (a) The product of two Gaussian PDFs is an unnormalized Gaussian, $\mathcal{N}(\mu_1, \Sigma_1)\mathcal{N}(\mu_2, \Sigma_2) = \eta\mathcal{N}(\mu, \Sigma)$, for some constant η. Give the expressions for μ and Σ in terms of μ_1, μ_2, Σ_1, and Σ_2.

   Solution:

   Starting by multiplying two PDFs

$$\begin{aligned}
&N(\mu_1, \Sigma_1)N(\mu_2, \Sigma_2) \\
=&\eta_1 \exp(\frac{1}{2}(x-\mu_1)\Sigma_1^{-1}(x-\mu_1))\eta_2 \exp(\frac{1}{2}(x-\mu_2)\Sigma_2^{-1}(x-\mu_2)) \\
=&\eta_1\eta_2 \exp\left[-\frac{1}{2}(x^T(\Sigma_1^{-1}+\Sigma_2^{-1})x - 2x^T(\Sigma_1^{-1}\mu_1+\Sigma_2^{-1}\mu_2) + \mu_1^T\Sigma_1^{-1}\mu_1 + \mu_2^T\Sigma_2^{-1}\mu_2)\right]
\end{aligned}$$

   Now, modify the normalization coefficient and compare terms with $N = \eta\exp(-\frac{1}{2}(x^T\Sigma^{-1}x - 2x^T\Sigma^{-1}\mu + \mu^T\Sigma^{-1}\mu))$:

$$\begin{aligned}
\Sigma^{-1} &= \Sigma_1^{-1} + \Sigma_2^{-1} \\
\Sigma^{-1}\mu &= \Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2 \\
\mu &= \Sigma(\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2) \\
&= (\Sigma_1^{-1} + \Sigma_2^{-1})^{-1}(\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2)
\end{aligned}$$

   Therefore, the product of two Gaussian PDFs is another Gaussian PDF with:

$$\mu = (\Sigma_1^{-1} + \Sigma_2^{-1})^{-1}(\Sigma_1^{-1}\mu_1 + \Sigma_2^{-1}\mu_2), \Sigma = (\Sigma_1^{-1} + \Sigma_2^{-1})^{-1}$$

   (b) $\mathbb{E}[(X-\mu)(X-\mu)^T] = \mathbb{E}[XX^T] - \mu\mu^T$, where $\mu := \mathbb{E}[X]$. Hence $\mathrm{Cov}(X)$ can be written in either of these two forms.

   Solution:

$$\begin{aligned}
E[(X-\mu)(X-\mu)^T] &= E[XX^T - \mu X^T - x\mu^T + \mu\mu^T] \\
&= E[XX^T] - E[\mu X^T] - E[X\mu^T] + E[\mu\mu^T] \\
&= E[XX^T] - \mu E[X^T] - E[X]\mu^T + \mu\mu^T \\
&= E[XX^T] - \mu\mu^T - \mu\mu^T + \mu\mu^T \\
&= E[XX^T] - \mu\mu^T
\end{aligned}$$

   the 3rd equation holds true because μ is a constant vector $E[\mu] = \mu$.

(c) If two RVs X and Y are independent, they are uncorrelated, $\text{Cov}(X, Y) = 0$. (Note: The converse is not true in general, but is true for jointly Gaussian RVs. But you don't have to show this.)

Solution:

$$\begin{aligned}
\text{Cov(X,Y)} &= E\left[(X - \mu_X)(Y - \mu_Y)^T\right] \\
&= E\left[XY^T\right] - \mu_X \mu_Y^T \\
&= \int xy^T p(x,y) dxy - \mu_x \mu_y^T \\
&= \int xy^T p(x)p(y) dX dY - \mu_x \mu_y^T \\
&= \int\int xp(x) dx y^T p(y) dy - \mu_x \mu_y^T \\
&= \int \mu_x y^T p(y) dy - \mu_x \mu_y^T \\
&= \mu_x \int y^T p(y) dy - \mu_x \mu_y^T \\
&= 0
\end{aligned}$$

The 4th equation is true because X and Y are independent, $p(x, y) = p(x)p(y)$.

3. For a Gaussian random vector $X \sim \mathcal{N}(\mu, \Sigma)$, we call the region $\mathcal{B}(\epsilon) = \{x \mid p(x) \geq \epsilon\}$ the error ellipse of $\mathcal{N}(\mu, \Sigma)$.

   (a) Show that $\mathcal{B}(\epsilon)$ defines an ellipse in $\mathbb{R}^n$ by expressing it in a quadratic form in x with a positive definite matrix.

   Solution:

   For any $x \in \mathcal{B}(\epsilon)$, we must have

   $$\frac{1}{\sqrt{(2\pi)^d|\Sigma|}} exp(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)) \geq \epsilon.$$

   Thus if we define $a = \epsilon\sqrt{(2\pi)^d|\Sigma|}$, we must have

   $$\begin{aligned} \exp(-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)) &\geq a, \\ \frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu) &\leq -\ln(a), \\ (x-\mu)^T\Sigma^{-1}(x-\mu) &\leq -\ln(\epsilon^2(2\pi)^d|\Sigma|), \end{aligned}$$

   where the last inequality defines the error ellipse in $\mathbb{R}^n$. Thus, since Σ is positive definite, we must have Σ^{-1} positive definite; i.e., $\mathcal{B}(\epsilon)$ defines an ellipse.

   (b) For the 2D case, show that $\epsilon = \frac{1-P}{2\pi|\Sigma|^{1/2}}$ gives the error ellipse with the property $Pr\big(X \in \mathcal{B}(\epsilon)\big) = P$ for a given probability $0 < P < 1$. Hint: Use the parameterization $w = \Sigma^{-1/2}(x-\mu)$ that we saw in class to convert the Gaussian to a standard form, then integrate.

   Solution:

   From part (a), we have, for $x \in B(\epsilon)$,

   $$(x-\mu)^T\Sigma^{-1}(x-\mu) \leq -\ln(\epsilon^2(2\pi)^2|\Sigma|).$$

   Let us introduce a change of variables, $\omega = \Sigma^{-\frac{1}{2}}(x-\mu)$.
   Then, we have:

   $$\begin{aligned} \omega^T\omega &\leq -ln(\epsilon^2(2\pi)^2|\Sigma|), \\ d\omega &= |\Sigma^{-\frac{1}{2}}|dx. \end{aligned}$$

   We can compute $Pr\big(X \in \mathcal{B}(\epsilon)\big)$ by integrating $p(x)$ over all $x \in \mathcal{B}$,

   $$\begin{aligned} Pr\big(X \in \mathcal{B}(\epsilon)\big) &= \int_{\mathcal{B}(\epsilon)} p(x)dx, \\ &= \int_{B(\epsilon)} \frac{1}{2\pi\sqrt{|\Sigma|}} \exp(-\frac{1}{2}\omega^T\omega)|\Sigma|^{\frac{1}{2}}d\omega, \\ &= \frac{1}{2\pi}\int_{B(\epsilon)} \exp(-\frac{1}{2}\omega^T\omega)d\omega. \end{aligned}$$

   We then convert the integral to polar coordinates, where, for $\omega = [\omega_1, \omega_2]$, we let $r = ||\omega||, \theta = \texttt{atan2}(\omega_1, \omega_2)$, yielding

   $$\begin{aligned} Pr\big(X \in \mathcal{B}(\epsilon)\big) &= \frac{1}{2\pi}\int_0^{2\pi}\int_0^{(-ln(\epsilon^2(2\pi)^2|\Sigma|))^{\frac{1}{2}}} \exp(-\frac{1}{2}r^2)rdrd\theta \\ &= \frac{1}{2\pi}2\pi(-e^{-\frac{1}{2}r^2})\Big|_0^{(-ln(\epsilon^2(2\pi)^2|\Sigma|))^{\frac{1}{2}}} \\ &= \frac{1}{2\pi}2\pi(-e^{-\frac{1}{2}r^2})\Big|_0^{-ln(1-P)^{2\,\frac{1}{2}}} \\ &= P \end{aligned}$$

where the 2nd equation converts the integration to polar coordinates and the fourth equation substitutes ϵ with $\frac{1-P}{2\pi|\Sigma|^{\frac{1}{2}}}$

4. Write a script (in Matlab, Julia, or Python) to plot the 2D error ellipse you found above with $P = 0.95$, given input parameters (μ, Σ). Plot 2D error ellipses overlying 1000 samples drawn from the distribution for three different 2D Gaussians of your choice. Attach your plots. Hint: The Matlab functions `randn` and `sqrtm` will come in handy, Julia functions `randn` and `sqrt(A::AbstractMatrix)`, or Python functions `numpy.random.randn` and `scipy.linalg.sqrtm` will come in handy.

   Solution:

   Code solutions are attached. Our code yields the following, for some random values of μ, Σ, Note that, indeed, roughly 19/20 samples appear to fall within the error ellipse.

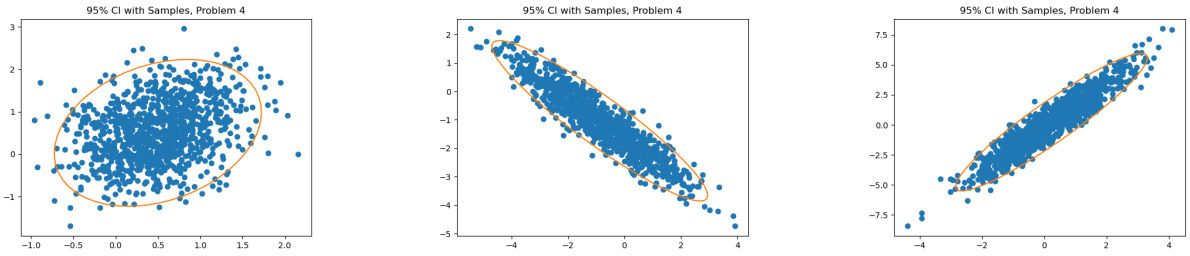


Figure 1: Gaussian error ellipses and samples for three different parameter values.

Code for HW solution:

```
import numpy as np
from matplotlib import pyplot as plt
import scipy.linalg

def plot_ellipse(P, mu, Sigma, label="", color="C1"):
    """
    Plots an error ellipse in R2 with probability P,
    for a Gaussian distribution with mean mu, covariance Sigma.

    Args:
        P: confidence interval to plot, scalar in [0, 1].
        mu: Mean of Gaussian, numpy array in R^2.
        Sigma: Covariance of Gaussian, numpy array in R^{2x2}.
        label: (optional) label for the pyplot line.
        color: (optional) color for the pyplot line.
    """
    # Compute radius of "error circle" in normalized coords.
    r = np.sqrt(-2*np.log(1-P))

    # Compute values for border of "error circle."
    theta = np.linspace(0, 2*np.pi)
    w = np.stack((r*np.cos(theta), r*np.sin(theta)))

    # Apply linear transform to un-normalize coordinates.
    x = scipy.linalg.sqrtm(Sigma)@w + mu

    plt.plot(x[0,:], x[1,:], label=label, c=color)

def plot_noise(mu, Sigma, N=1000):
    """
    Samples points from a 2D Gaussian and scatters them on the current figure.

    Args:
        mu: mean of Gaussian, numpy array.
        Sigma: covariance of Gaussian, numpy array.
        N: number of samples to draw and scatter.
    """
    samps = scipy.linalg.sqrtm(Sigma) @ np.random.normal(size=(2, N)) + mu
    plt.scatter(samps[0,:], samps[1,:])

def pb4():
    """
    Script for Problem 4. Draws 1k samples from a given Gaussian, and
    scatters them over a 95% error ellipse for the distribution.
    """
    mu = np.random.randn(2,1)
    Sigma = np.random.randn(2,2)
    Sigma = Sigma.T @ Sigma

    plot_ellipse(0.95, mu, Sigma)
    plot_noise(mu, Sigma)
    plt.title("95% CI with Samples, Problem 4")
```

```
if __name__ == "__main__":
    # Generate three plots of random Gaussians.
    for ii in range(3):
        pb4()
        plt.figure()

    plt.show()
```